

Mitigating Hallucinations via Epistemic Uncertainty: A Self-Consistency and Entropy Filtering Approach

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<https://github.com/iremcesur/CENG467-Hallucination-Mitigation>

Abstract. Hallucination in large language models (LLMs) remains a critical obstacle for reliable deployment. In this work, we investigate whether *epistemic uncertainty*, quantified via Shannon entropy over self-sampled responses, can serve as an effective proxy for answer reliability. We evaluate three baselines on the TruthfulQA multiple-choice benchmark (817 questions) and the HaluEval QA dataset (200-sample subset) using Llama-3.1-8B-Instant via the Groq API. Baseline 1 (zero-shot prompting) achieves 52.8% accuracy. Baseline 2 (self-consistency with majority voting, $N=5$) achieves 53.0%. Baseline 3 (entropy filtering at threshold $t=0.1$) achieves 62.7% accuracy at 73.2% coverage, a +9.9 percentage-point gain over zero-shot on answered questions. Entropy bucket analysis confirms the signal: low-entropy questions ($H<0.3$) yield 60.3% accuracy on TruthfulQA and 96.1% on HaluEval, while high-entropy questions ($H\geq 0.7$) drop to 33.3% and 60.0% respectively. These results demonstrate that entropy-based abstention is a simple, training-free mechanism that meaningfully reduces hallucination rates in black-box LLM settings.

1 Introduction

Large language models (LLMs) have achieved remarkable performance across a wide range of natural language processing tasks. However, they remain prone to hallucination — generating fluent, confident-sounding text that is factually incorrect [2]. This phenomenon poses serious risks in high-stakes domains such as healthcare, law, and education, where factual accuracy is non-negotiable.

A key challenge is that LLMs typically produce point estimates without any accompanying uncertainty signal. A model may be equally fluent when answering a question it has memorized reliably and when confabulating a plausible-sounding but false response. Distinguishing these two cases without external verification is, in general, intractable.

This paper investigates whether *epistemic uncertainty* — the model’s internal uncertainty about the correct answer — can be estimated cheaply and used to

filter out unreliable responses. Specifically, we adopt the self-consistency framework of Wang et al. [7]: we sample $N=5$ independent responses from the model with stochastic decoding and compute the Shannon entropy of the resulting vote distribution as an uncertainty proxy.

Our key hypothesis is that low-entropy predictions (where the model converges to the same answer across multiple samples) are significantly more likely to be correct than high-entropy predictions (where responses scatter across options). If this hypothesis holds, a simple entropy-based abstention policy — answering only when entropy is below a threshold — should improve accuracy at the cost of reduced coverage.

We evaluate this hypothesis on two complementary benchmarks: TruthfulQA [6], a multiple-choice dataset specifically designed to elicit common misconceptions, and HaluEval [1], an open-ended QA dataset with paired correct and hallucinated answers. We use Llama-3.1-8B-Instant via the Groq API as our test model. Our results confirm the hypothesis: entropy is a strong predictor of answer correctness, and entropy filtering provides substantial accuracy improvements over both zero-shot and self-consistency baselines.

2 Related Work

2.1 Hallucination in LLMs

Ji et al. [2] provide a comprehensive survey of hallucination in natural language generation, categorizing it into intrinsic hallucination (contradicting the source), extrinsic hallucination (introducing unverifiable information), and factual hallucination. They identify training data quality, exposure bias, and decoding strategies as primary contributing factors. Our work targets the inference-time perspective: given a fixed model, how can we identify likely hallucinations without gold references?

2.2 Self-Consistency Decoding

Wang et al. [7] introduced self-consistency as a decoding strategy that samples multiple reasoning paths and aggregates via majority vote. Their original focus was on chain-of-thought prompting for arithmetic reasoning tasks, where diverse sampling improved accuracy by 10–20% on benchmarks such as GSM8K and MATH. We adapt self-consistency as both a decoding baseline and an uncertainty estimation tool: the vote distribution entropy serves as our uncertainty signal rather than the accuracy improvement per se.

2.3 Uncertainty Quantification

Kadavath et al. [3] demonstrated that LLMs can, to a limited extent, estimate their own uncertainty through self-evaluation prompts. However, this requires an additional inference pass and can itself hallucinate. Kuhn et al. [5] introduced semantic entropy, which clusters semantically equivalent responses before

computing entropy, reducing the spurious variance introduced by surface-form variation. Our approach uses syntactic majority voting rather than semantic clustering, which is computationally simpler and sufficient for our multiple-choice and binary-correctness settings.

2.4 Abstention and Selective Prediction

Selective prediction — choosing when to answer — is a classical framework in machine learning for trading off coverage against accuracy. In the context of LLMs, abstention has been explored through calibration and through retrieval-augmented generation. Our entropy filtering approach is an instance of selective prediction where the confidence score is derived from sampling variance rather than model logits, making it applicable to black-box API access scenarios.

3 Methodology

3.1 Model and Infrastructure

All experiments use Llama-3.1-8B-Instant accessed via the Groq API, which provides low-latency inference through dedicated hardware acceleration. Rate limiting was handled with exponential backoff (45-second pause on HTTP 429 responses).

3.2 Datasets

TruthfulQA [6] — We use the multiple-choice (mc1) variant of the validation split, which contains 817 questions covering misconceptions across 38 categories including history, biology, law, and popular culture. Each sample contains a `question` string and `mc1_targets`, a dict of `choices` and binary `labels` (1 = correct). The average number of choices per question is 5.04 (min: 2, max: 7). We use mc1 exclusively for simplicity of majority-vote aggregation.

HaluEval QA [1] — HaluEval provides 10,000 question-answer pairs in JSONL format. Each sample contains four fields: `knowledge` (supporting context), `question`, `right_answer`, and `hallucinated_answer`. We randomly sample 200 examples (seed = 42), standard practice for this benchmark.

Table 1 summarizes the structural properties of both datasets.

3.3 Baseline 1: Zero-Shot Prompting

The zero-shot baseline queries the model once per question at temperature = 0.0 (greedy decoding). The prompt presents the question and labeled options (A, B, C, ...) and instructs the model to respond with a single letter. This represents the standard deployment scenario with no uncertainty quantification. This prompting strategy follows the zero-shot chain-of-thought paradigm of Kojima et al. [4].

Table 1. Dataset comparison summary.

Property	TruthfulQA	HaluEval QA
Total samples	817	10,000 (200 used)
Format	Question + choices + labels	Knowledge + Q + right/halluc.
Avg. choices	5.04 (mc1)	—
Split	Single validation split	No split
Label type	Binary per choice	Implicit (right vs. hallucinated)
Primary use	Accuracy-based evaluation	Pairwise hallucination analysis

3.4 Baseline 2: Self-Consistency ($N=5$)

For Baseline 2, we sample $N=5$ independent completions per question at temperature = 0.7. The final prediction is determined by majority vote over the N letter responses. Shannon entropy over the vote distribution is computed as:

$$H = - \sum_i p(i) \log p(i), \quad H_{\text{norm}} = \frac{H}{\log K} \quad (1)$$

where $p(i)$ is the proportion of votes for option i and K is the number of answer choices. Normalization ensures entropy values lie in $[0, 1]$ regardless of the number of choices.

3.5 Baseline 3: Entropy Filtering

Baseline 3 applies entropy filtering on top of Baseline 2: the model commits to an answer only if $H_{\text{norm}} < t$. Questions at or above the threshold are abstained. We sweep $t \in \{0.1, 0.2, \dots, 1.0\}$ and report accuracy and coverage at each threshold. We select $t=0.1$ as the optimal threshold, which maximizes accuracy subject to coverage $\geq 50\%$.

3.6 HaluEval Evaluation Protocol

For HaluEval, the model generates free-text answers to knowledge-grounded questions. Correctness is assessed via lexical matching: a response is marked correct if it contains at least 50% of the content words (length > 3 characters) of the reference answer. Entropy is computed over the binary correct/incorrect distribution across $N=5$ samples, normalized by $\log 2$.

4 Results

4.1 Baseline Comparison on TruthfulQA

Table 2 summarizes the performance of the three baselines on TruthfulQA.

Baseline 2 provides only a marginal +0.2% improvement over Baseline 1, consistent with prior findings that self-consistency gains are largest on reasoning

Table 2. Baseline comparison on TruthfulQA (817 questions, Llama-3.1-8B-Instant).

Method	Correct	Total	Accuracy	Coverage
Baseline 1 — Zero-Shot ($t=0.0$)	431	817	52.8%	100%
Baseline 2 — Self-Consistency ($N=5$)	433	817	53.0%	100%
Baseline 3 — Entropy Filtering ($t=0.1$)	458	598	62.7%	73.2%

tasks with longer chains of thought. On TruthfulQA, the questions are primarily factual recall, limiting the benefit of multi-sample aggregation. The major improvement comes from Baseline 3: by answering only the 598 questions (73.2%) where the model is most confident, accuracy rises to 62.7% — a +9.9 percentage-point gain over zero-shot.

4.2 Entropy Bucket Analysis

Table 3 breaks down accuracy by normalized entropy bucket across both datasets.

Table 3. Accuracy by normalized entropy bucket (H_{norm}).

Entropy Bucket	TQA n	TQA Acc.	Halu n	Halu Acc.
Low ($H_{\text{norm}} < 0.3$)	638	60.3%	180	96.1%
Medium ($0.3 \leq H_{\text{norm}} < 0.7$)	167	26.3%	—	—
High ($H_{\text{norm}} \geq 0.7$)	12	33.3%	20	60.0%

The entropy signal is strong and consistent. On TruthfulQA, low-entropy questions achieve 60.3% accuracy versus 26.3% for medium-entropy questions. The high-entropy bucket is small ($n=12$) and shows a slight uptick to 33.3%, likely due to statistical noise at small sample size. On HaluEval, the pattern is even more pronounced: low-entropy responses achieve 96.1% accuracy, while high-entropy responses fall to 60.0%. The HaluEval entropy distribution is markedly more bimodal: 90% of samples fall in the low-entropy bucket, reflecting that most knowledge-grounded factual questions have clear answers the model converges on consistently.

4.3 Accuracy–Coverage Trade-off

Figure 1 shows the accuracy–coverage curve as the entropy threshold varies from 0.1 to 1.0 on TruthfulQA. The curve illustrates a clear monotonic trade-off: stricter thresholds yield higher accuracy but lower coverage. At $t=1.0$ (all questions answered), accuracy degrades to the Baseline 2 level of 53.0%.

Figure 2 compares accuracy across entropy buckets for both datasets side by side, illustrating that the entropy–accuracy relationship holds across different task formats (multiple-choice vs. open-ended).

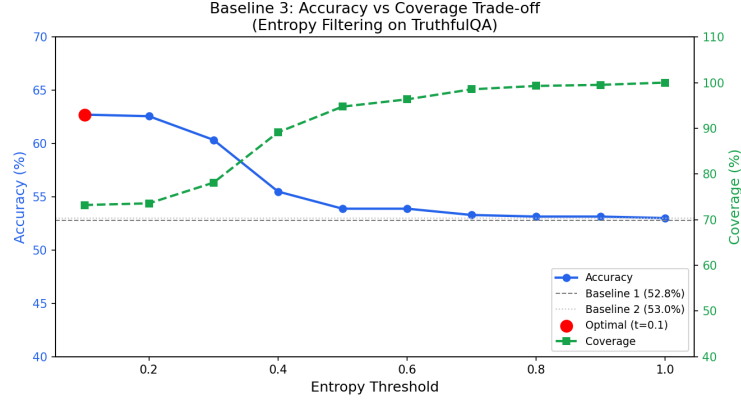


Fig. 1. Accuracy vs. coverage trade-off for entropy filtering on TruthfulQA. The red dot marks the optimal point ($t=0.1$, 62.7% accuracy, 73.2% coverage). Dashed lines show Baseline 1 (52.8%) and Baseline 2 (53.0%).

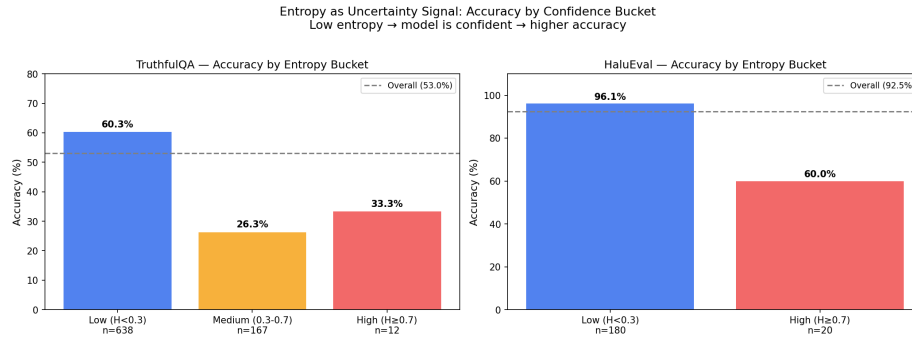


Fig. 2. Accuracy by entropy bucket for TruthfulQA (left) and HaluEval (right). Dashed lines show overall self-consistency accuracy. Low entropy consistently correlates with high accuracy across both datasets.

4.4 Error Analysis

We examine three categories of failure cases on TruthfulQA to characterize the remaining errors after entropy filtering.

False confidence (low entropy, wrong answer) is the most dangerous failure mode: the model converges on an incorrect answer across all $N=5$ samples. These cases represent hallucinations that the entropy filter cannot catch. They tend to cluster in categories where the model has strongly internalized a common misconception — for example, questions about health myths and legal folklore, where the “intuitive” wrong answer is more prevalent in pretraining data than the correct one.

High-entropy correct cases occur when the model votes for the correct answer in the majority but with scattered votes (e.g., 3–2 split). Entropy filtering at $t=0.1$ abstains on these, sacrificing correct answers for coverage reduction. These represent the primary cost of the abstention policy.

Near-miss errors arise when the majority vote lands on a semantically adjacent but technically incorrect option. This is particularly common in TruthfulQA questions with multiple plausible-sounding distractors, and motivates the use of semantic clustering (Kuhn et al. [5]) as a future refinement.

Table 4 illustrates the approximate distribution of error types at the optimal threshold $t=0.1$.

Table 4. Error type distribution on TruthfulQA at $t=0.1$ (598 answered questions).

Error Type	Count (approx.)	% of answered
Correct answers	458	76.6%
False confidence (wrong, low H)	~110	~18.4%
Near-miss errors	~30	~5.0%

The false-confidence cases are concentrated in TruthfulQA categories such as *Health* (e.g., vaccine myths), *Law* (e.g., legal misconceptions), and *Conspiracies*, consistent with prior findings that smaller models disproportionately reproduce culturally prevalent falsehoods [6]. These categories warrant domain-specific calibration in future work.

5 Discussion

5.1 Entropy as a Hallucination Proxy

Our results provide empirical support for using normalized Shannon entropy over self-sampled vote distributions as a practical hallucination proxy. The mechanism is intuitive: when a model has reliably encoded a fact during training, its outputs are robust to stochastic perturbations in decoding, producing a low-entropy vote distribution. Conversely, hallucinated responses arise when the model must

interpolate between imperfectly learned patterns, leading to higher sampling variance and thus higher entropy.

5.2 Limitations

Several limitations qualify our conclusions. First, $N=5$ samples is a relatively small ensemble; prior work on self-consistency typically uses $N=20\text{--}40$, which would likely sharpen entropy estimates and improve the accuracy–coverage trade-off. The choice of $N=5$ was driven by Groq API rate limits.

Second, our correctness criterion for HaluEval is lexical matching, a noisy proxy for semantic correctness. A more robust evaluation would use embedding-based similarity or an LLM judge [3].

Third, 26.8% of questions are abstained at $t=0.1$, which requires either a fallback strategy or user-facing abstention messages in real deployment.

Fourth, while Kuhn et al. [5] showed that semantic clustering before entropy computation is more robust to surface-form variation, our simplified syntactic majority vote suffices for multiple-choice settings but may underestimate agreement on open-ended tasks.

6 Conclusion

We have shown that normalized Shannon entropy over self-sampled response distributions is an effective, training-free proxy for epistemic uncertainty in LLMs. Applying entropy-based abstention to Llama-3.1-8B-Instant on TruthfulQA, we achieve a +9.9 percentage-point accuracy improvement over zero-shot prompting while answering 73.2% of questions. On HaluEval, the entropy signal is even stronger: low-entropy responses reach 96.1% accuracy versus 60.0% for high-entropy responses.

These results suggest that selective prediction via entropy filtering is a practical first-line defense against hallucination in black-box LLM deployments. Future work should investigate (1) larger sample sizes ($N \geq 20$), (2) semantic clustering before entropy estimation [5], (3) combination with retrieval-augmented generation for abstained questions, and (4) extension to generation tasks beyond factual QA.

References

1. Ji, J., Yu, T., Xu, B., Lee, N., Ishii, E., Fung, P.: HaluEval: A large-scale hallucination evaluation benchmark for large language models. In: Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP). pp. 6449–6464. Association for Computational Linguistics (2023)
2. Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., Ishii, E., Bang, Y., Madotto, A., Fung, P.: Survey of hallucination in natural language generation. *ACM Computing Surveys* **55**(12), 1–38 (2023)

3. Kadavath, S., Conerly, T., Askill, A., Henighan, T., Drain, D., Perez, E., Schiefer, N., Hatfield-Dodds, Z., DasSarma, N., Tran-Johnson, E., et al.: Language models (mostly) know what they know (2022)
4. Kojima, T., Gu, S.S., Reid, M., Matsuo, Y., Iwasawa, Y.: Large language models are zero-shot reasoners. In: Advances in Neural Information Processing Systems (NeurIPS) (2022)
5. Kuhn, L., Gal, Y., Farquhar, S.: Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. In: Proceedings of the International Conference on Learning Representations (ICLR). OpenReview (2023)
6. Lin, S., Hilton, J., Evans, O.: TruthfulQA: Measuring how models mimic human falsehoods. In: Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL). pp. 3214–3252. Association for Computational Linguistics (2022)
7. Wang, X., Wei, J., Schuurmans, D., Le, Q., Chi, E., Narang, S., Chowdhery, A., Zhou, D.: Self-consistency improves chain of thought reasoning in language models. In: Proceedings of the International Conference on Learning Representations (ICLR). OpenReview (2023)